Repairing the Negation Blind Spot in Hebrew Text Embeddings

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The Negation Blind Spot

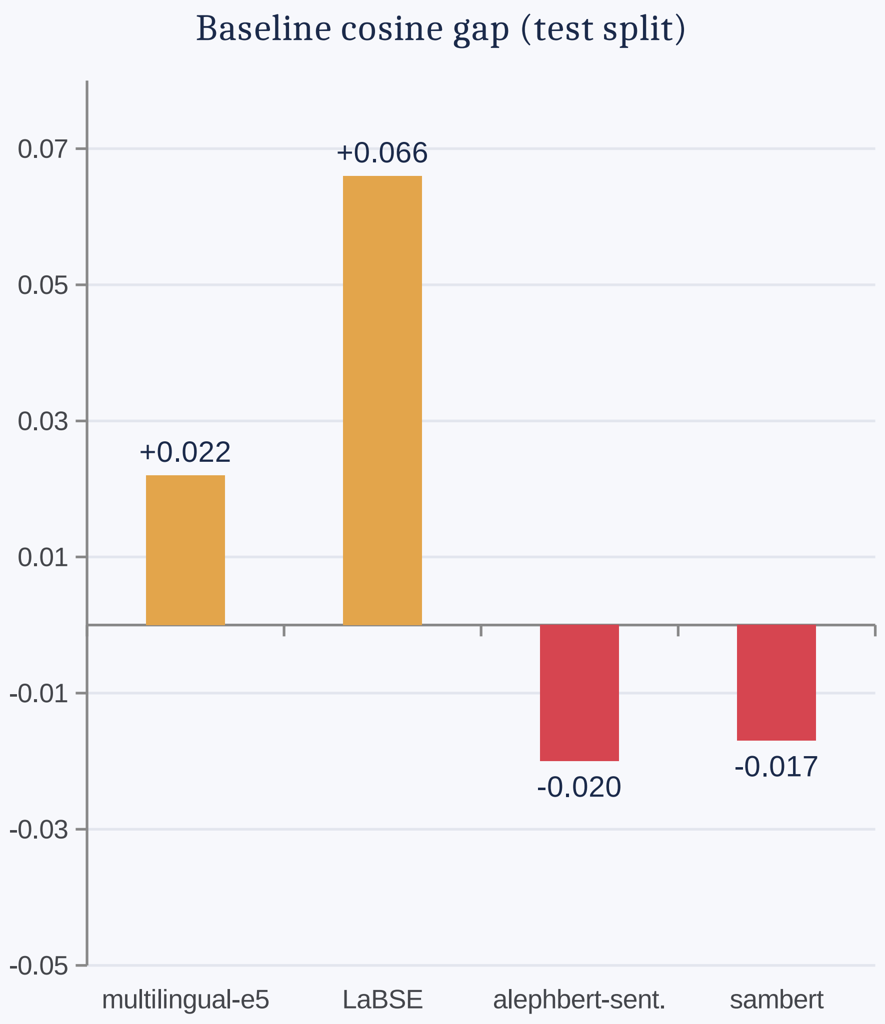
A sentence and its negation share almost every word — but mean the opposite thing.

`cosine gap = cos(target, paraphrase) − cos(target, negation)`

A negation-sensitive model should score this well above zero.

2 of 4

frozen Hebrew embedding models score a sentence's negation as ≥ its paraphrase — on their own test



Three Questions

1

Repair it post-hoc?

Fix a frozen embedder’s vector space directly — no retraining.

2

Edit vs. fuse?

Compare a representation-space fix against blending in an external NLI signal.

3

Is the fix real?

Or can a metric “improve” by degenerate means — collapsing the space until everything looks alike?

(3) turned out not to be hypothetical — it shaped both interventions.

Related Work

Diagnosing it

NevIR (Weller+ 2024), CONDAQA (Ravichander+ 2022): retrieval & QA models rank negation pairs near chance.

Editing the space

Hard-debiasing removes a direction (Bolukbasi+ 2016); activation steering adds one (Turner+, Zou+ 2023). We amplify the sentence's own signal — neither.

Fusing a signal

Cross-encoder re-ranking (NevIR) inspired nli_rerank: blend cosine with an external NLI judgment.

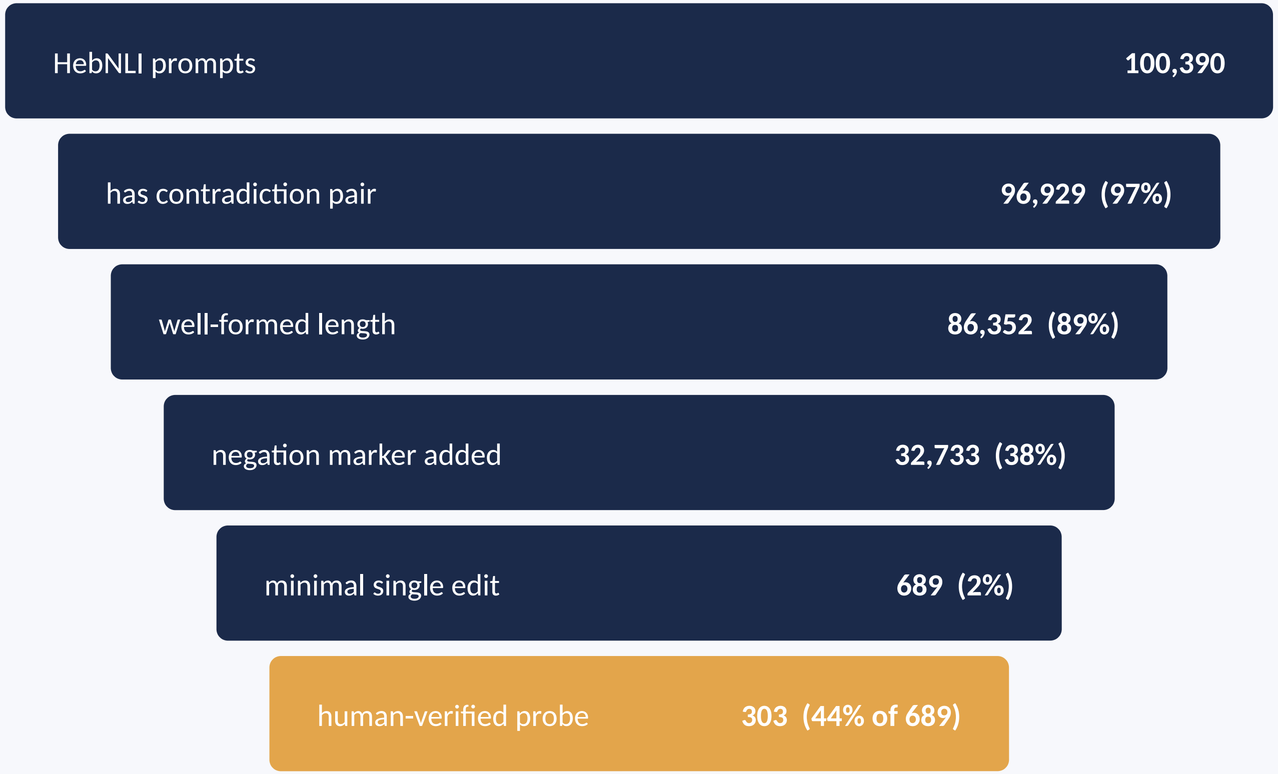
Hebrew resources

AlephBERT (Seker+ 2022) and HebNLI underlie both our embedder and our fine-tuned classifier.

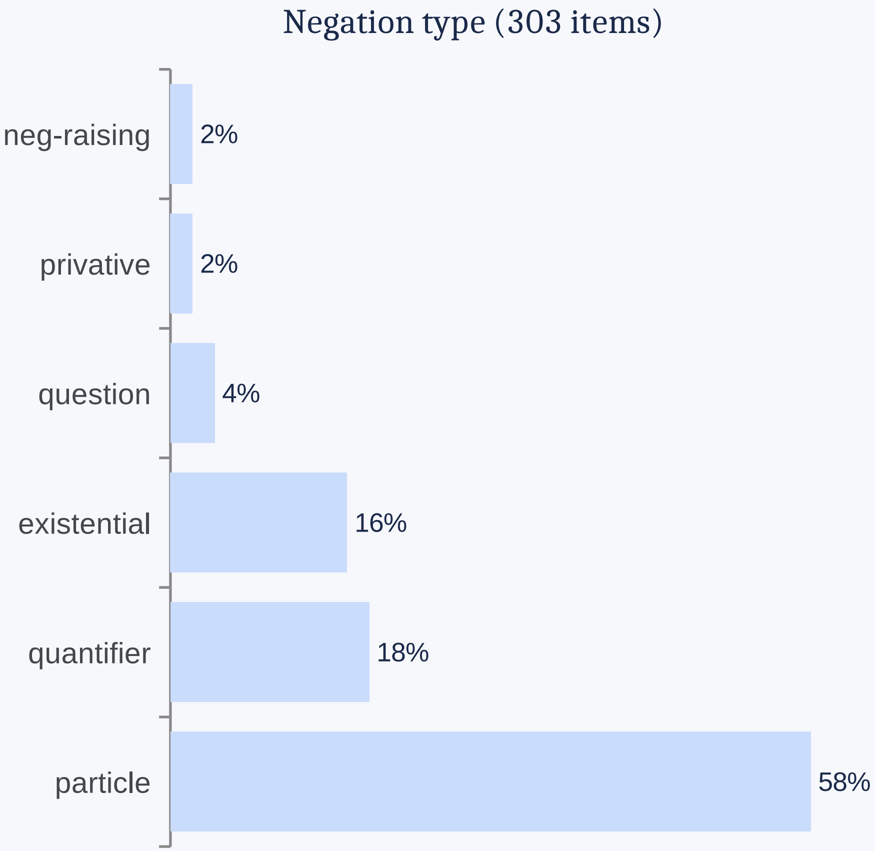
No prior work measures or repairs this specifically for Hebrew embeddings.

Building a Hebrew Negation Probe

Mined from HebNLI — not sampled from “contradiction”, hand-verified that the opposition is really negation.



62% of HebNLI contradiction pairs contain no negation marker at all.



152 / 151 train / test split — stratified so no type sits entirely on one side.

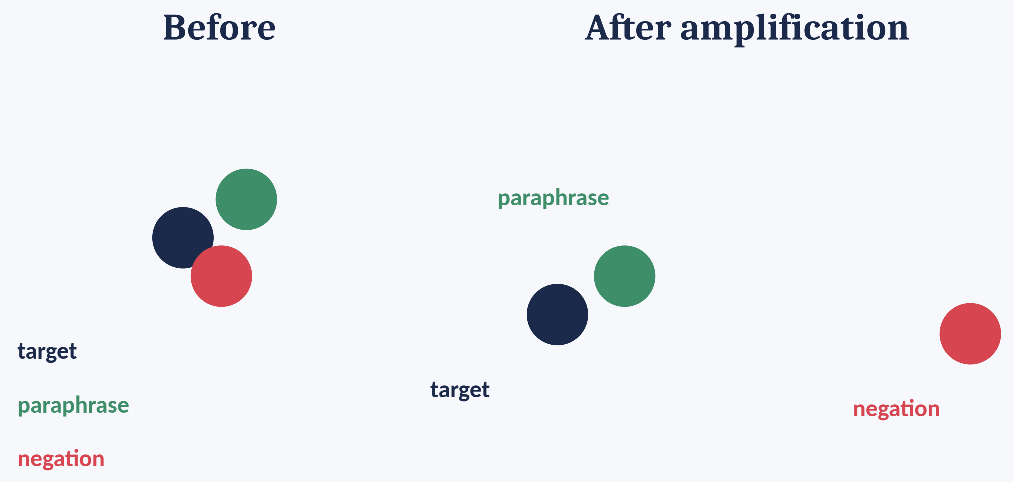
Fix #1 — projection: edit the space

Estimate a negation direction d, then amplify each sentence's own component along it.

$$v' = v + (\gamma - 1) \cdot ((v \cdot d) - \mu) \cdot d$$

$\gamma = 1$	identity — no change
$\gamma > 1$	amplifies the negation signal already in the sentence
d	mean-difference, or a logistic-regression hyperplane normal

Deliberately not debiasing: removing d would make a sentence and its negation MORE alike — the opposite of the goal.



Selecting γ

5-fold CV on train, argmax cosine gap — never touches test.

⚠ Gap-only selection turned out to be unsafe — next slide.

Fix #2 — nli_rerank: fuse a signal

Leave the embedding alone. Blend its cosine with a Hebrew NLI classifier's judgment.

```
combined = (1−λ)·cos(a,b) + λ·[P(entail) − P(contra)]
```

Classifier

AlephBERT fine-tuned on HebNLI — 79.6% test accuracy (macro-F1 79.4%).

Decontamination

Probe items' 689 promptIDs held out of fine-tuning — else the classifier would have memorized the probe.

Choosing λ

21-point grid per model. Max pairwise accuracy among λ whose Hebrew-STS-dev Spearman stays within 0.02 of λ=0.

Locked once

Selected on train/STS-dev, evaluated exactly once on test/STS-test.

`λ = 0`

pure cosine (baseline)

`λ = 1`

pure NLI — embedder discarded entirely

selected λ per model:

0.05 – 0.35

a little NLI goes a long way

Both Fixes Need a Second Guard

Maximizing the target metric alone finds the same degenerate optimum in either intervention.

PROJECTION — γ

$\gamma = 1000$

sim_unrelated

0.975

guarded (γ = 12): sim_unrelated 0.350

NLI_RERANK — λ

$\lambda = 1$

STS Spearman

0.53

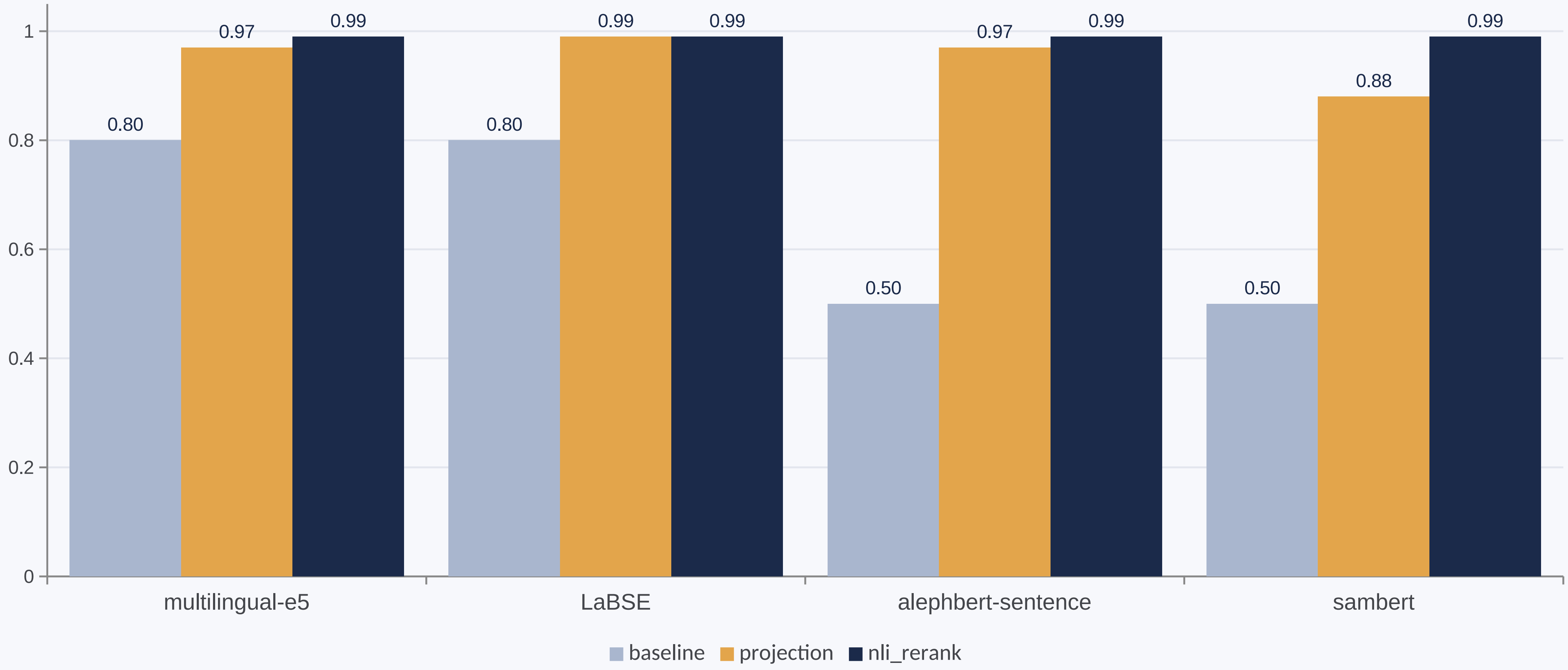
guarded (λ = 0.05–0.35): STS moves ≤ 0.07

Same fix either way: keep only the values an independent guard metric approves.

Held across all 4 models — the raw metric keeps "improving" right up to collapse.

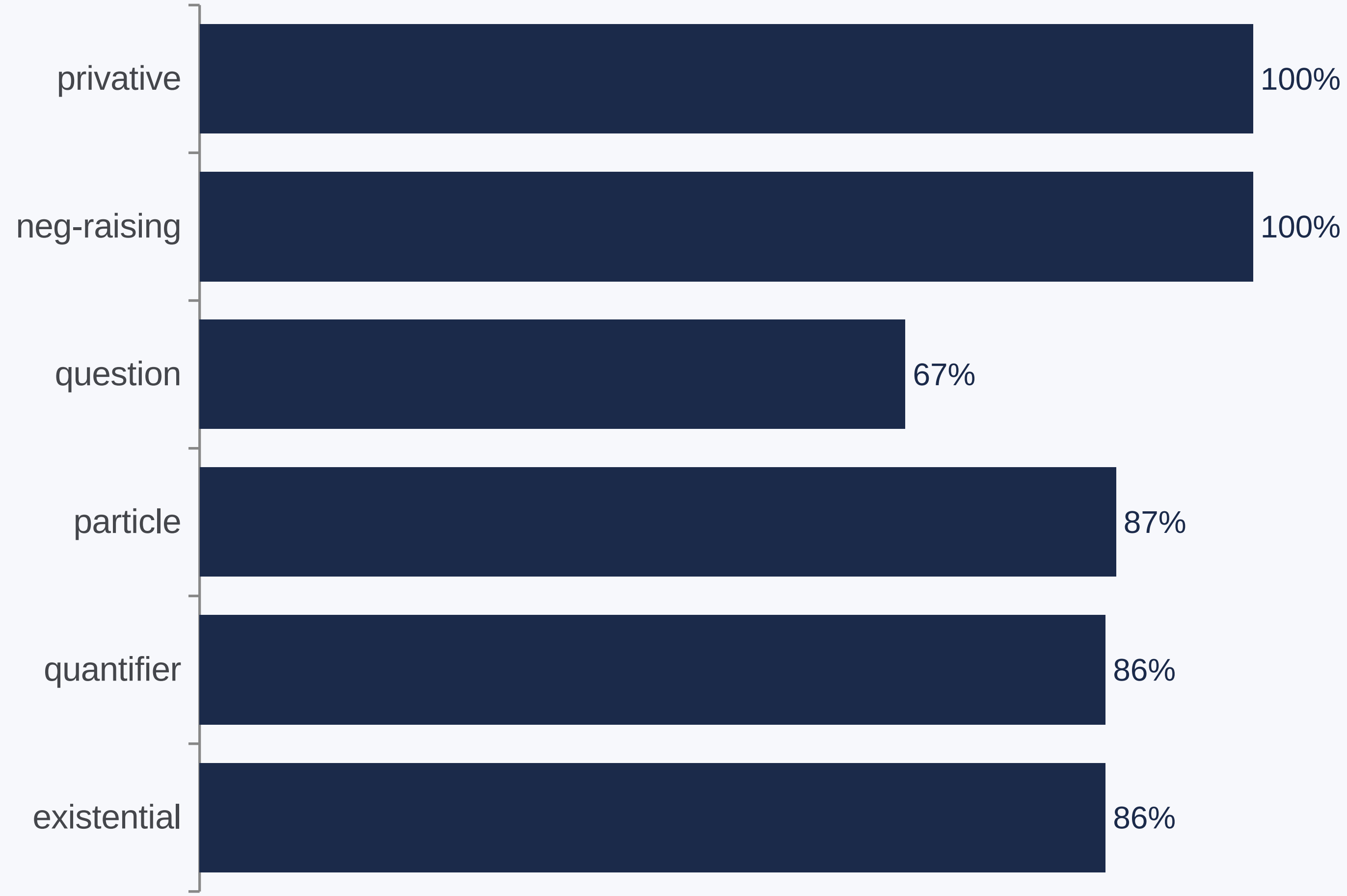
Head-to-Head: Pairwise Accuracy

Chance = 0.50. Both fixes help; nli_rerank wins on every model, with a real STS guard instead of a proxy.



Where projection Still Struggles

Share of the baseline gap closed, by negation type (aggregated across all 4 models).



Biggest 3 categories (89% of test items)

close 86–87% of their gap

privative & neg-raising

close 100% — but n=12 each (3 per model). Plausibly a small-n ceiling effect, not genuine ease.

Contribution & Limitations

The methodological finding is the contribution:

Amplifying a concept direction safely needs an independent guard metric — gap-only selection reliably finds the degenerate optimum, and cross-validation does not protect against it.

Limitations

- 303-item probe — smallest categories (n=3–4/model) are suggestive, not conclusive
- sim_unrelated is a proxy, not a real similarity benchmark
- Hebrew STS-B is a translation of English STS-B, not natively annotated
- multilingual-e5's collapse guard is unresolved within this search grid

Future Work

- Wire a real Hebrew STS guard into projection's γ-selection
- Extend the probe with a 4th sentence → true 0.25-chance NevIR metric
- Investigate why multilingual-e5 resists the collapse-safe region

Thank You

Code, data, results, and Colab notebooks:

`github.com/ItayBoros/hebrew-negation-embeddings`

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